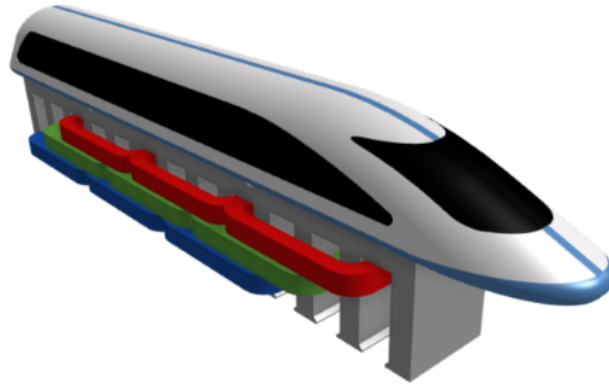


Mini Maglev: Precise Speed Control of a Linear Induction Motor



Ada Li
Amy Li
Cyrus Young
Dhruv Arun

Project Sponsors:
Engineering Physics Project Lab

Project Code: 2504
Course Code: ENPH 459

Department of Engineering Physics
University of British Columbia
June 10, 2025

Executive Summary

The capstone project, sponsored by the UBC Engineering Physics Project Lab, focused on designing a miniature linear induction motor (LIM). Our goal was to achieve precise thrust control using field-oriented control (FOC), representing the initial phase toward a fully functional mini-Maglev. The project enabled students to learn advanced electromechanical systems through a scalable and hands-on platform that integrated power electronics, control systems, and electromagnetism.

Throughout the year, our team successfully designed and built an LIM test platform and partially implemented a real-time FOC using a STM32 microcontroller. The system featured an Integrated Circuit gate driver, a 3-phase inverter to generate controlled AC waveforms, real-time current sensing, and full execution of Clarke and Park transforms along with space vector PWM. These components enabled us to demonstrate stable closed-loop speed control—a key milestone that validates the feasibility of precise LIM thrust control.

Although stable magnetic levitation was beyond the current scope, our work this year prepared the technical foundation for the next stage of the project. Achieving stable thrust and levitation will require further FOC algorithm development and electromagnetic simulations to refine stator geometry and force control. This project confirms the viability of the LIM-FOC architecture and provides a clear, actionable road-map: move forward with modeling and control refinements, transitioning from simple thrust-only motion to fully stabilized Maglev operation next year.

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1 Introduction

1.1 Sponsor Information

Our capstone is generously sponsored by the UBC Engineering Physics Project Lab consisting of Dylan Gunn, Miti Isbasescu, and Gregory Reid.

1.2 Background and Significance

Maglev trains, or Magnetic Levitation trains, are a modern type of transport that has produced, so far, many of the world's fastest operational trains. Beyond its speed, the design and construction of a Maglev train involve input from many fields, requiring knowledge of electromagnetic physics, electronics, mechanical engineering, and control systems. Due to the wide variety of fields covered for the building of a Maglev, the project is ideal for a Capstone project, as it enables learning and developing skills in multiple areas of study. Obviously, building an actual size Maglev train is unrealistic, but building a miniaturized version of one (around the size of a chocolate bar) is feasible.

1.3 Problem Statement and Objectives

The objective is to build a miniaturized version of a Maglev train; basically, a small track with electromagnets to levitate and propel the train along the track. A PCB motor driver generates the necessary current, and a microcontroller is used to control the current (using a PI loop) to stabilize and move the train.

1.4 Scope and Limitations

This year's main objective is to build a linear motor and show that we can produce step changes in thrust. The design requires a stator + rotor test stand, a three-phase inverter, various state sensors, and an STM32 microcontroller running our developed FOC algorithms. Although complete stable magnetic levitation is the goal for next year, demonstrating precise thrust operation now ensures the system can deliver precise speed control to the train. The final mini-Maglev platform will serve both educational and demonstration purposes within the UBC Engineering Physics Project Lab, illustrating principles of electromagnetics, power electronics, and control theory.

2 Discussion

2.1 Outline of the Discussion (SLD)

Before diving into the details of the project, we would like to emphasize our objective and how it motivated aspects of our design. Our goal this year was to de-risk the design of a mini Maglev train by demonstrating that we can **precisely control the speed of a linear induction motor**.

As this project is relatively complex, we structure the discussion in a way that moves from theoretical content to practical implementation:

- A linear induction motor, in simple terms, is an electromechanical device. In order to understand and build one, our discussion will start with a review of the fundamentals of electromagnetism. Having established this physics background, we will then explain how thrust is generated in a linear induction motor, and how we can effect instantaneous changes in thrust necessary for precise speed control.
- To test our implementation of the FOC algorithm, we built an encoded test stand, as well as motor driver. The rest of the discussion describes the requirements and design of this hardware, the FOC software implementation, and the tests we have run on the system.

2.2 Theoretical Framework

2.2.1 Electromagnetism

The entire mechanics of the train itself, why it levitates and accelerates, can be explained by the physics of electromagnetism. The fundamental basis for why we can even build a train that magnetically levitates derives from Maxwell's equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (4)$$

Where $\mathbf{E}$, $\mathbf{B}$ are the electric and magnetic fields, μ_0 and ϵ_0 are the permeability and permittivity of free space, and $\mathbf{J}$ is the electric current density (the current per unit area). The third equation, Faraday's law, and the fourth equation, the Ampere-Maxwell law, together arguably describe the entire physics of the project. Faraday's law describes how a changing magnetic field will produce an induced electric field, while the Ampere-Maxwell law describes how magnetic fields are generated by changing electric fields and currents.

2.2.2 Producing Thrust

Another aspect of electromagnetism that is crucial is the Lorentz force law (with no electric field) shown below:

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad (5)$$

This equation describes why the track accelerates the train. As alternating current is generated in the coils of the track, a changing vertical magnetic field is generated. This generates a current in the train itself (called eddy currents), via Faraday's law. The train is thus accelerated as the current flowing within it feels a force generated by the magnetic field.

Now, for the test setup, we are using a rotor (effectively a disk that can spin freely), but the physics is the same: eddy currents are induced in the rotor, which in turn, interact with the traveling magnetic field, thus producing a force on the rotor.

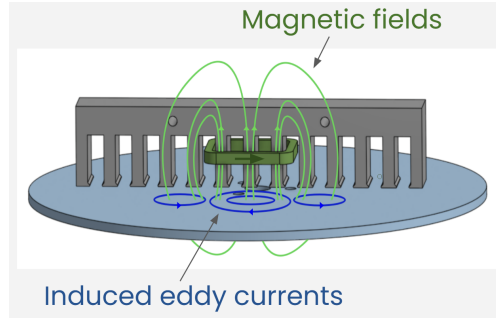


Figure 1: Induced rotor currents

2.2.3 Control Methods

The thrust generated by the linear induction motor can be expressed as $\tau \propto \psi_R I_R$ where τ is thrust, ψ_R is the rotor flux, and I_R is the rotor current [1].

Now, we can see two ways of changing the thrust of the linear induction motor: change the rotor flux, or change the rotor current. We cannot be changing the rotor flux if we want to effect step (instantaneous) changes in thrust. This is because magnetic fields store energy, and it takes a non-negligible amount of time to change the energy of a system. Instead, we look to change the rotor current.

Recall that in our linear induction motor, the rotor is simply a sheet of aluminum, and is not connected to any power supply. Therefore, the stator must induce currents in the rotor, as well as create the magnetic field. The problem then boils down to how can we control the stator current such that it increases its thrust component without changing its flux producing component?

Field-oriented control (FOC) aims to decouple torque and flux control. FOC manipulates two orthogonal current components—one to control flux, the other to control torque—independently and in real time.

d-q Axis Decoupling: The key insight behind FOC is to transform the three-phase stator currents $\{i_a, i_b, i_c\}$ into a rotating reference frame aligned with the rotor (or stator) magnetic field. In this rotating frame, the stator currents split into two orthogonal components:

- i_d : The direct-axis current that primarily controls the flux (or magnetizing component).
- i_q : The quadrature-axis current that directly controls torque (or thrust in a linear motor).

By maintaining a constant i_d for the desired flux level, and modulating i_q to control torque, we achieve independent control of flux and torque 2. An important note is that this requires a jump in the phase, frequency, and magnitude of the stator current phasor.

Clarke & Park Coordinate Transforms: To go from the motor's three-phase domain to the rotating reference frame, two mathematical transforms are used in sequence:

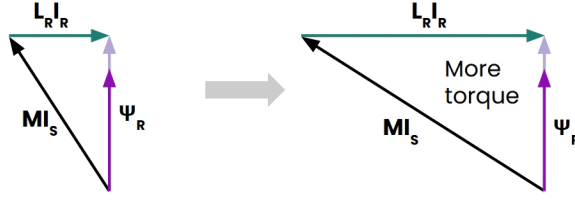


Figure 2: FOC (decoupled current control)

1. Clarke Transform (3-phase $\rightarrow$ 2-phase stationary frame)

- *Inputs:* i_a, i_b, i_c .
- *Outputs:* i_α, i_β , representing the stator current in a two-axis orthogonal system fixed in space.
- This transform is effectively a projection of the three-phase vectors onto two perpendicular axes (α and β).

2. Park Transform (2-phase stationary frame $\rightarrow$ 2-phase rotating frame)

- *Inputs:* i_α, i_β and the rotor flux angle θ .
- *Outputs:* i_d, i_q , where i_d is aligned with the rotor flux vector and i_q is perpendicular to it.
- By “locking” one axis (the d -axis) to the rotor flux, the flux-producing current is held constant, and the torque-producing current can be quickly varied.

Rotor Flux Angle θ : This angle represents the instantaneous position of the rotor’s magnetic field. Estimating or measuring it (via sensor or observer algorithm) is crucial for accurate FOC.

2.2.4 Approaches to Indirect Rotor Flux Estimation

The Field-Oriented Control principle relies on the correct alignment of the d -axis in the rotating reference frame with the north pole of the actual flux wave. Otherwise, then part of the q -current change we command will not be aligned with the actual non-flux-producing current component, so will be opposed by inductance. Then the torque will change slower than is possible. In Direct Rotor Flux Orientation, the flux wave is directly sampled by magnetic flux sensors, but adding these sensors adds a mechanical point of failure. Therefore we choose to pursue Indirect Orientation by estimating the rotor flux using the current measurements and a dynamical model of the stator and rotor fluxes.

From a review of textbooks ([1], [5], [6], [7]), it has been repeatedly emphasized that accurate knowledge of the motor parameters, particularly the rotor time constant ($\tau_r = \frac{L_r}{R_r}$) is crucial to accurate estimation of the rotor flux magnitude and phase angle. We are not confident that we can determine the rotor time constant to a few percentage points of error, because we are unsure how to characterize our rotor, which is a homogeneous conductor with no discrete windings, and because the rotor resistance will increase as the rotor heats up from ohmic losses.

Therefore, we will aim to use closed-loop observers of the rotor flux which are robust to large measurement error in the rotor time constant, and potentially adaptive online estimation of the time constant itself. Meth-

ods presented in our textbooks perform the estimation using the back-EMFs induced in the stator windings by the rotor flux wave, the stator currents, and the rotor speed. Trzynadlowski [6] presents an example observer of rotor flux acting as a Model-Reference Adaptive System (MRAS) (Figure 3) and simulations which demonstrate that the actual flux magnitude and angle can be accurately tracked even when our model's rotor time constant is off by 22%.

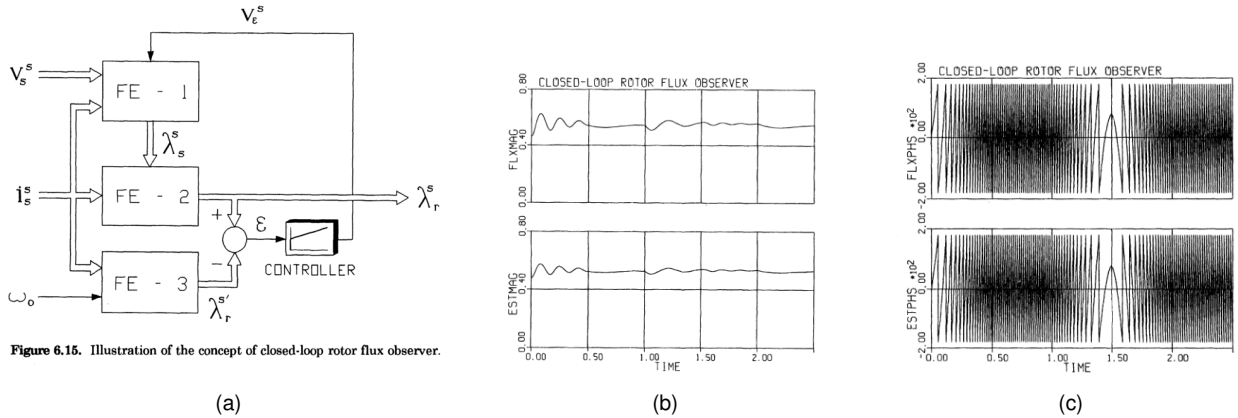


Figure 3: Concept of the MRAS Rotor Flux Estimator from [6]. Motor simulation produced plots of the correct flux magnitude (b) and phase angle (c) on top, and the values predicted by the rotor flux observer on the bottom, which accurately track the correct values. [6]

2.3 Linear Induction Motor Implementation

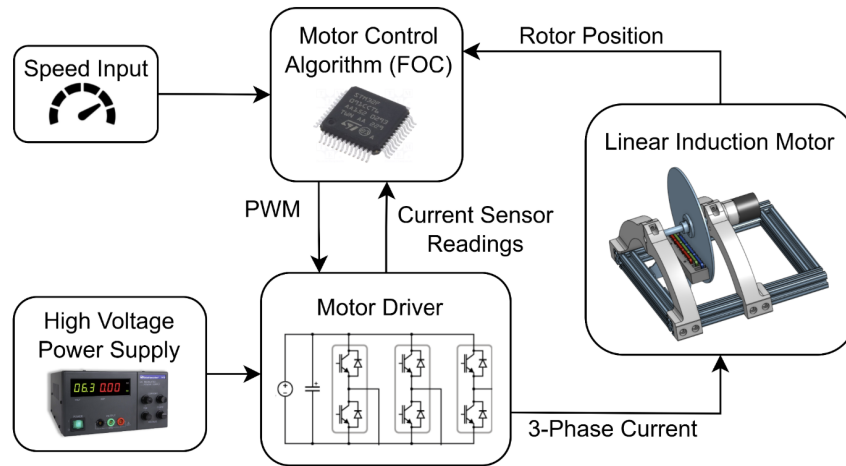


Figure 4: System level diagram

Shown in Figure 4 is the system level diagram of our project, which relates how the hardware (linear induction motor test stand and motor driver) interface with our motor control algorithm. In this section, we will walk through the system level diagram as we describe important design decisions in our hardware and software implementation that help us meet the objective of precise speed control.

2.3.1 Linear Induction Motor Test Stand

A regular Maglev (stator) moves along a stationary track (rotor). However, to test the speed control of our linear induction motor more easily, we opted to fix the stator and allow the rotor to rotate (see Figure 5) to essentially prevent us from "running out of track." An optical encoder with 2400 ticks per revolution and a max speed of 5000 rpm is attached to the rotor to allow us to measure rotor position and speed. The stator consists of 20 AWG wire with 3 poles per phase and 20 turns per coil (Figure 6). These specifications were largely based on the prototype in [2].

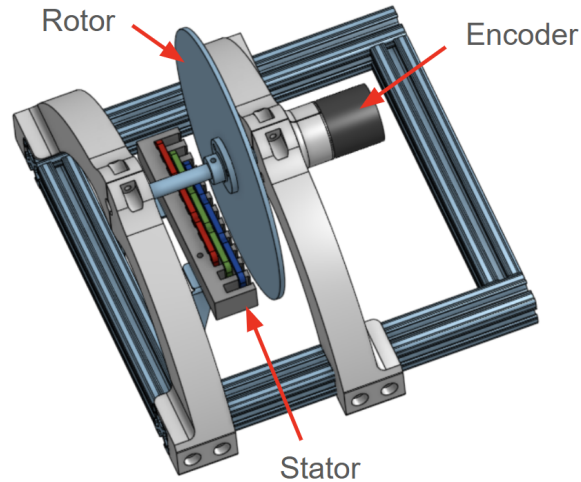


Figure 5: Test stand

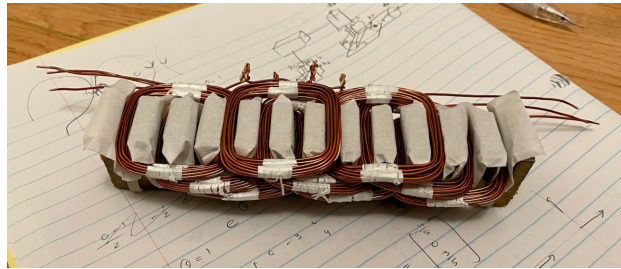


Figure 6: Stator construction

2.3.2 Motor Driver

Recall that to instantaneously change the thrust output of the linear induction motor, we must be able to instantaneously change the stator current. Due to coupling between the stator and the rotor, the effective transient inductance is small [1], but still requires us to apply a high voltage if we want to change the stator current quickly. Therefore, the inverter was designed to withstand up to 100 V and 15 A continuously.

PWM switching frequency was a second consideration for the driver. To minimize audible switching noise and current ripple, we would ideally drive the motor at a PWM frequency of 10 to 20 kHz. This required us to select optocouplers and gate drivers capable of operating at this frequency.

As well, we wanted to completely isolate the microcontroller from the inverter circuit due to the noisy nature

of the inverter. This meant that not only did we need optocouplers between the microcontroller PWM outputs and the gate drivers, we also needed to isolate the current measurement signals. We accomplished this by using isolation amplifiers based on Sigma-Delta capacitive isolation. Rough current measurement specifications are given below.

Specification	Notes
Current range	We selected 1 m Ω shunt resistors which allow us to measure up to 40 A of current. To adapt to different current ranges, we also have 2-5 m Ω shunt resistors that can easily be swapped in. The tradeoff is between current resolution and range.
Noise	From limited scope readings, the current measurements have about 4 mA of high frequency noise which can and should be low pass filtered out in the future (see Figure 9 for an example of current sensor noise).
Accuracy	The current measurements have an offset that needs to be calibrated out. Minus this, the error of the current measurements have up to 5% error (from limited tests where we ran currents through power resistors).

Table 1: Current sensor specifications

We took care while routing the PCB to avoid stray capacitances and inductances, for example, by separating signal traces and power traces by ground planes, keeping traces short and physically separating noise sensitive components (see Figure 7).

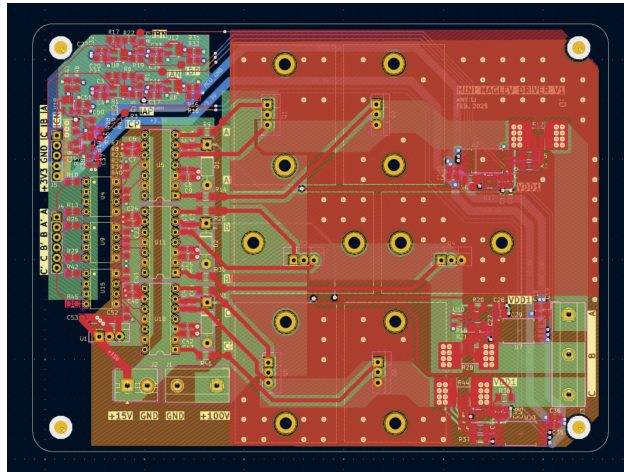


Figure 7: Motor driver PCB

2.3.3 Microcontroller Selection

Precise speed control requires us to run a control loop at ideally 20 kHz, which dictated our choice of microprocessors. We had two microcontrollers on hand to experiment with: the STM32F767ZI on Nucleo-144 development board, and the ESP32-WROOM-32. Both are fast (216 MHz and up to 240 MHz clock speed respectively) and have enough GPIO pins and peripherals to support for motor control. It took a while to learn how to use the STM32 toolchain, but following it turned out to be easy to program as the Hardware Access Library and device peripherals are both thoroughly documented, and the graphical code-generation

tool STM32CubeMX provides easy initialization of peripherals like timers, ADCs, and the Direct Memory Access controller (DMA) without diving into low-level driver code. The ESP-32 does not have such easy low-level control of peripherals but its motor control libraries have sufficiently detailed configuration options to implement standard algorithms just as well. Finally, both the STM32 and ESP32 support the wealth of debugging utilities from SEGGER, including RTT Viewer for debug printing and J-Scope for live plotting of global variables.

The most relevant difference for our application is the nonlinearity and poorer stability of the ESP32's internal ADCs. Without an external ADC module, the STM32 is the better choice for our closed-loop current control requiring current sensing.

2.3.4 FOC Software

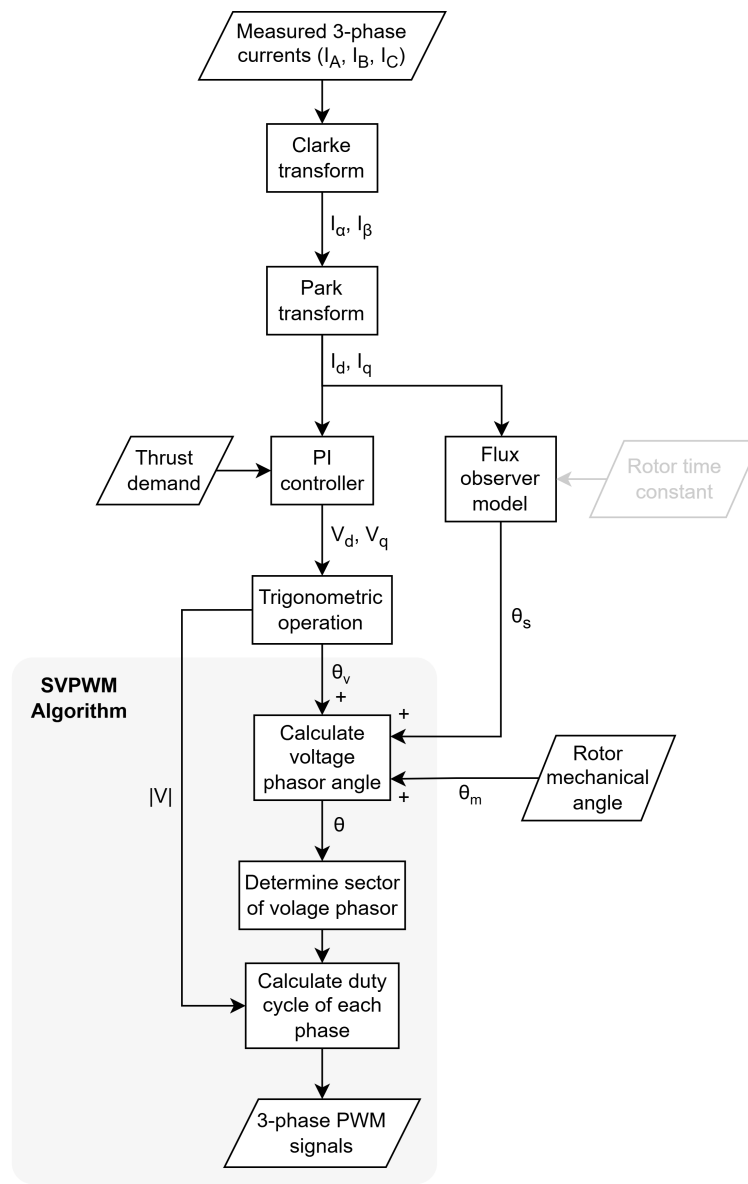


Figure 8: FOC software implementation flowchart

To remind the reader, FOC allow us to have step changes in thrust that would not be possible with simpler control algorithms. Because of the complexity of implementing FOC, we broke up our approach into steps:

1. First, we wanted to verify that we could output the 3-phase sinusoidal PWM signals necessary to drive the inverter. On the STM32, the advanced-control timer TIM1 can generate 3-phase complementary PWM signals with manually set dead-time insertion which prevents shoot-through currents in the MOSFETs. By sinusoidally varying the duty cycles of the three channels with a 120° phase offset, we could drive the rotor at up to 3500 rpm steady speed while commanding 1000 Hz electrical frequency and drawing 4 A from a lab power supply. Thrust was small using this limited 35 V supply.
2. Next, we sought to implement a variable frequency drive (VFD), which is essentially FOC at steady state. The VFD uses a SVPWM algorithm (based on [4]) which is outlined in the grey box in Figure 8. It takes the magnitude and angle of a voltage vector as inputs, and outputs 3-phase sinusoidal PWM signals. (See Appendix B for a detailed description.) With VFD, we also successfully drove the rotor. The measured sinusoidal currents can be seen in Figure 9. Note that this was run in a 1 kHz control loop on an ESP32 for quick experimentation; increasing the control loop frequency results in smoother sinusoids.
3. FOC requires that we can read the stator current sensors and rotor encoder.

We used three channels on ADC1 to read our current sensors. The ADCs have a dynamic range of 0 to 3.3V with 12-bit resolution. The ADC can be set to perpetually scan our three current sensor readings and transfer them to a circular buffer via the Direct Memory Access controller, which does not rely on the CPU. Therefore we can write non-blocking code which can take advantage of ADC sensor readings at a rate of 600 kHz (calculated in Appendix A), which can be used for oversampling/averaging. The current sensor readings can be live-plotted by storing them in a global variable and accessing them from J-scope. Logging the currents on the timescale of the inverter switching requires implementing RTT communication in code, which we still need to do.

Finally, the general purpose timers have the internal logic to directionally read a quadrature encoder. On the STM32 TIM[x] is used for this purpose. We can compute the speed by dividing the number of pulses in a known time interval precisely given by one of the internal timers.

4. As mentioned previously, knowing the rotor flux angle is crucial to FOC, since we use this angle to decouple and separately control the thrust and magnetizing components of the stator current. However, without magnetic field sensors in our setup, we must indirectly estimate the rotor flux angle from the stator currents. It can be shown [3] (see also Appendix B) that in steady state, $\theta_r = \theta_m + \int_0^t \frac{R_r}{L_r} \frac{I_d}{I_q} dt$, where θ_r is the rotor flux angle, θ_m is the rotor mechanical angle (as measured by the encoder¹), R_r is the rotor resistance, L_r is the rotor inductance, and I_q and I_d are the stator q and d axis currents, respectively.

A major issue, unfortunately, is that we cannot directly measure the resistance and inductance of the rotor, being a plate of aluminum. The next best alternative, is to experimentally determine the rotor time constant ($\frac{L_r}{R_r}$) by setting the desired d and q stator currents equal, and varying our estimate of

¹Do not be deceived by the word "mechanical." This is actually an electrical angle, and therefore we cannot directly use the encoder angle as the rotor mechanical angle. We need to perform a conversion, which is outlined in more detail in Appendix B

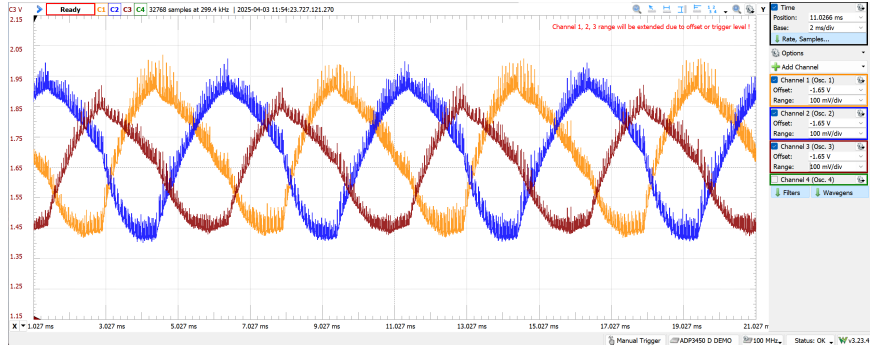


Figure 9: Measured sinusoidal stator currents

the time constant in our Field-Oriented thrust controller [2] (shown by the greyed out input in Figure 8). The correct rotor time constant would then give the maximum thrust.

We did implement the entire FOC algorithm on the stm32 MCU. However, due to rushed integration near the end of this project, we were not able to successfully find this rotor time constant. Next steps and suggestions for this are included in the next section.

2.4 System Summary

Having described our system in detail, we feel that is appropriate to wrap up the discussion with our detailed system level diagram in Figure 10.

3 Work for Next Year

3.1 Measuring the Rotor Time Constant

The main symptom that the rotor time constant experiment was not working was that the computed voltage phasor did not rotate in a way that would generate sinusoidal currents. Several immediate fixes (that we simply ran out of time to implement) could have resolved this:

1. One of the authors realized retrospectively that the FOC algorithm was running in a loop that was deliberately frequency limited, when it should have been placed elsewhere in the code and run as fast as possible.
2. Current measurements have an offset and noise that needs to be removed.
3. The experiment of finding the rotor time constant by looking for maximum thrust production essentially requires that the rest of the FOC implementation is correct. However, as can be seen in Figure 8, we have PI controller which we were not able to tune, since we did not have a running system that we could use to tune it. The problem then is one of guessing PI gains that could lead to a reasonable response.
4. We used a potentiometer as an input to change the rotor time constant. However, it is possible that the range of the potentiometer (0 to 1000) does not come close to the order of magnitude of the real

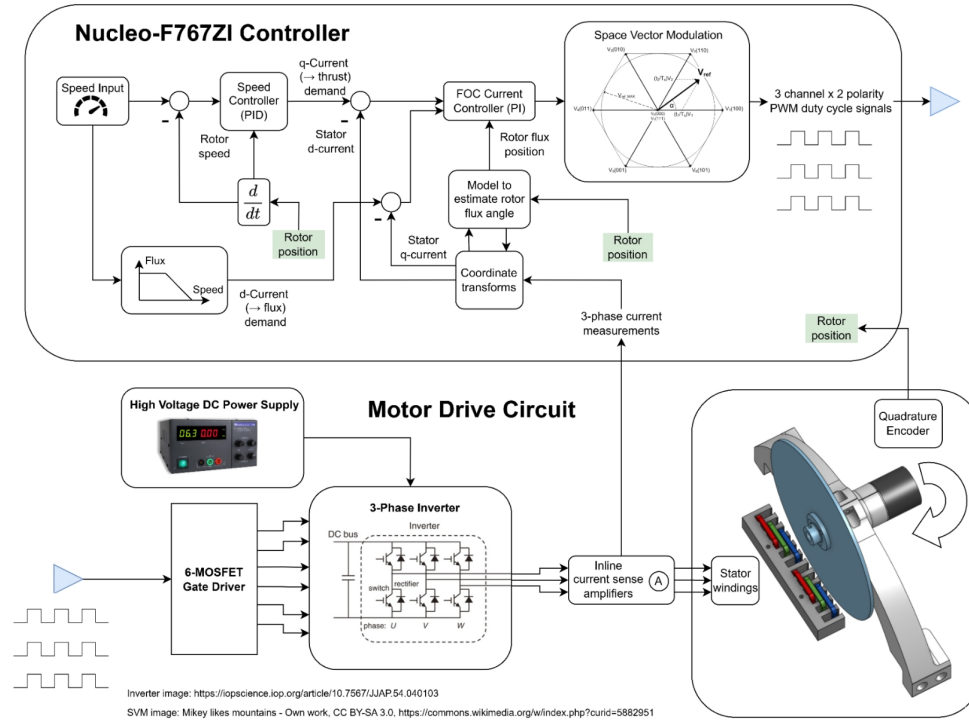


Figure 10: Detailed system level diagram

rotor time constant. We need to experiment with a larger range of values.

5. We made a lot of estimations when converting between encoder angle and rotor mechanical angle (in electrical degrees). We should double check this conversion, perhaps by running the motor at synchronous speed.
6. We need a better test setup to measure thrust. Our last-minute setup involved attaching a spring to the rotor and observing max displacement. However, the thrust of the motor while powered from the 35 V 4.5 A supply is very small. We need a more sensitive thrust measurement, and to hook the motor up to the 80 V 20 A supply.

3.2 Stator Optimization for Levitating Maglev Train

The current stator (Figure 6) measures about 20 x 20 x 140 mm and weighs about 300 grams, is too large to place on a miniature train model and too heavy to levitate. To reduce size and weight, we will:

1. Use smaller gauge wire (22-26 AWG) with the windings since we do not plan to attain the full current loading of 15A.
2. Make the stator skinnier. This will also benefit us by reducing the dominance of end effects, which make it difficult to characterize the stator flux-producing properties.
3. Consider removing the iron core, which would increase the air gap reluctance opposing the stator flux but could be compensated for by driving more current at higher frequency.

4. We may possibly benefit from the compactness and consistency of a PCB forcer. Compactness becomes increasingly important if we are to remove the iron core, as in a wound-wire stator, different phases will see a drastically different reluctance due to varying air gaps.

The motor driver PCB (Figure 7) must also fit onto the train. The largest space-users at present are the power MOSFETs and heat sinks, which we can downsize in the next PCB iteration by accepting lower current limits.

Due to the small length scale, the stator inductance might be quite small and so the stator currents might need to be driven at a very high frequency to produce optimum torque.

Specific implementations of levitation will also require more consideration; some approaches are outlined in Appendix A.

4 Conclusions

In this term, we showed that our motor drive PCB and MCU-based control system is fundamentally sound for driving an induction motor with a linear stator at an inverter frequency of 20 kHz. In addition, the non-negligible thrust that we were able to generate with a low power supply suggests that our more powerful supply will have enough juice (perhaps with some large capacitors) to allow us to drive rapid speed changes. This work is thus easily extensible: The drive circuitry is fundamentally the same no matter the control system and whether the motor is rotary or linear; we need only miniaturize the stator and on-board drive circuitry so it can be mounted on a miniature model train. The remaining hard part for motor control is in EM and controls simulation, software implementation, and measurement.

We believe that FOC control of a linear induction motor propelled train for optimum thrust is feasible, and we have identified the design work. Although the Field-Oriented Control principle without direct flux sensing critically relies on stable indirect estimation of the flux (stator, air gap, OR rotor) even with measurement error in the motor parameters, we can follow standard textbook methods which produce flux magnitude and angle errors much smaller than the measurement error of the rotor time constant, and we can use an online estimator of the rotor time constant to quickly reduce that error further.

We have not considered in great detail whether stable magnetic levitation is both possible and compatible with stable thrust control (as the stator air gap needs to be kept relatively constant). It is one of our goals next year to conduct a design study to ensure that our train and rail design would be compatible with adding magnetic levitation before building the rail. In any case, the project scope in implementing FOC for a non-levitating train is already a challenging and worthwhile enough problem that we will be satisfied even if we don't get to magnetic levitation.

5 Deliverables

1. Test stand CAD models
2. Motor driver PCB, simulations, scope traces
3. STM32 sinusoidal PWM test code

4. ESP32 VFD test code
5. STM32 FOC code
6. MATLAB Simulink Code

A Appendix A: Derivations and Formulas

A.1 Space Vector Pulse-Width Modulation (SVPWM)

Space Vector Pulse-Width Modulation (SVPWM) is a technique for generating three-phase PWM signals that more efficiently utilize the DC bus voltage compared to conventional sinusoidal PWM.

In the context of our Mini Maglev system, SVPWM is a key component of the Field-Oriented Control (FOC) architecture. Once the FOC algorithm has determined the required stator voltage vector $\vec{V}_{ref}$ -based on our desired d - and q -axis currents- SVPWM converts that reference to the appropriate switching commands for the three-phase inverter. By choosing the two nearest active states and inserting appropriate dwell times in each PWM cycle, the algorithm reconstructs $\vec{V}_{ref}$. This leads to lower harmonic distortion in the stator currents and reduces thrust ripple in the rotor, both important when striving for smooth, precise speed and thrust control.

Additionally, SVPWM yields better bus-voltage utilization than simpler modulation schemes. This is an advantage when the motor driver supply voltage is constrained, but rapid thrust changes are still required.

Our design requires switching at 20 kHz. Based on our evaluation of the timing constraints of our micro-controller and gate drivers, we conclude that SVPWM's real-time calculations can be supported. With the STM32's advanced-control timers and direct memory access (DMA) for current sensing, we achieve the necessary throughput, enabling the controller to update the PWM signals on every cycle.

A.2 Encoder Angle to Rotor Mechanical Angle Conversion

The rotor mechanical angle must be in units of electrical degrees to be of use in the FOC loop. In general, 360 electrical degrees is equivalent to one wavelength of the sinusoidal current in space, or simply the length of two poles in our linear induction motor. Therefore, to convert from the encoder angle to electrical degrees, we first eyeballed the average radius r of the rotor that the stator sees. This then gives us the distance $s = r\theta$ that the rotor moves for a given encoder angle. We then divide s by the length of two poles, and then multiply by 2π to get electrical degrees.

Evidently, this method is not extremely accurate: we estimate an average radius, but in reality, the system might not be as simple since the linear stator sees a different radius of the circular rotor at each point.

A method that we touched upon earlier was to run the motor at no load (synchronous speed), where we would expect very little slip. Then, we can compare the known flux wave speed with the speed recorded by the encoder to verify our estimate of the rotor mechanical angle conversion factor.

A.3 Derivation of Rotor Flux Observer Model

Several methods for obtaining a rotor flux observer exist, many which are quite complicated and involve state-space modeling. It turns out that if we assume a constant rotor flux, we can apply a simplified model to estimate the rotor flux angle, outlined in [3]. We will go over a partial derivation here.

A reminder of the notation before we begin:

From the equivalent circuit model of an induction motor, we get that $\frac{d\psi_r}{dt} = \frac{R_r}{L_r} L_m I_d - \frac{R_r}{L_r} \psi$. The steady state solution to this ODE is $\psi_r = L_m I_d$. We can leverage this since the goal of FOC is to keep the rotor flux constant (i.e., at steady state).

One can also show that slip speed $\omega_s = \frac{L_m R_r}{\psi_r L_r} I_q$. Substituting $\psi_r = L_m I_d$, we get that $\omega_s = \frac{R_r}{L_r} \frac{I_q}{I_d}$.

Finally, since $\omega_r = \omega_m + \omega_s$, we can integrate to get $\theta_r = \theta_m + \int_0^t \frac{R_r}{L_r} \frac{I_q}{I_d} dt$.

A.4 ADC Sampling Rate Calculation

Our ADC on the STM32767ZI is clocked off the PCLK2 timer divided by 4 (108MHz/4 = 27MHz). Scanning three channels at 12-bit resolution and converting their results takes $3 \times 15 = 45$ clock cycles. Therefore the sampling frequency is 27MHz/45 = 0.6MHz.

A.5 Design for Magnetic Levitation

There are two categories of magnetic levitation systems [8]. Electromagnetic suspension uses active control of attraction between electromagnets on the train hooking underneath permanent magnets in the rail. Electrodynamic suspension relies on repulsion between the train's stator and induced magnetic flux wave in the guide rail, which becomes strong enough to lift the train beyond a minimum speed. We will use the electrodynamic suspension approach for our rail design for two reasons:

1. Electrodynamic suspension is dynamically stable as the train sits in a repulsive potential well, whereas electromagnetic suspension is naturally unstable and would require us to get right another very responsive control system.
2. The guide rail geometry for electromagnetic suspension will be very hard to control over long stretches of track and costly to build. The guide rails for electrodynamic suspension are much simpler and do not require permanent magnets.

θ_r	rotor flux angle
θ_m	rotor mechanical angle
ω_r	rotor flux speed
ω_m	rotor mechanical speed
ω_s	slip speed
R_r	rotor resistance
L_r	rotor inductance
I_q	stator q-axis current
I_d	stator d-axis current
ψ_r	rotor flux

Table 2: Notation

3. We would naturally see levitation as a byproduct of thrust production with a miniaturized the stator on the bottom of our train.

We also need a way to measure the Maglev speed as it moves without being able to attach a physical encoder. We can use a linear strip + optical encoder combination, an indirect observer which uses the stator currents and induced back-emfs to indirectly estimate the train speed, or a combination of encoder, observer, and IMU for lowest uncertainty.

Finally, we need a way to deliver DC supply voltage to the train. A small onboard Li-Po battery could provide energy for a few minutes running but possibly at unacceptably low voltage and space + weight cost. For external options, we will consider a third rail or overhead wire system.

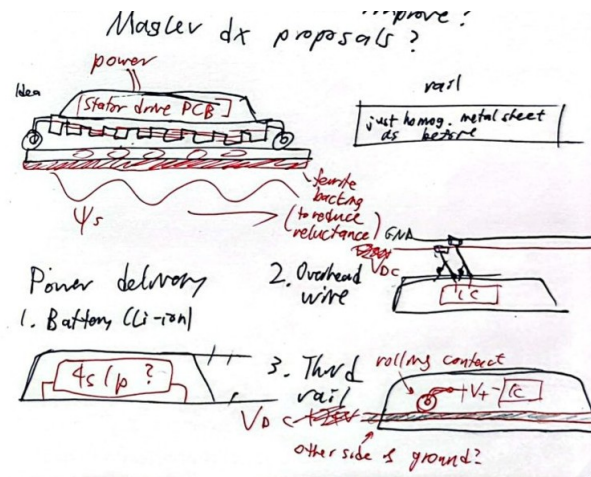


Figure 11: Options for power delivery to Maglev train

B Appendix B: Supplemental Materials

B.1 Circuit Schematics

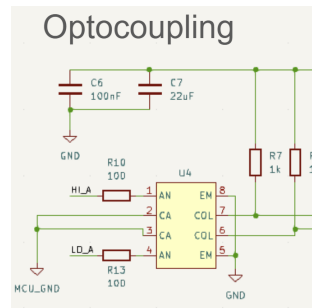


Figure 12: PCB of Optocoupler

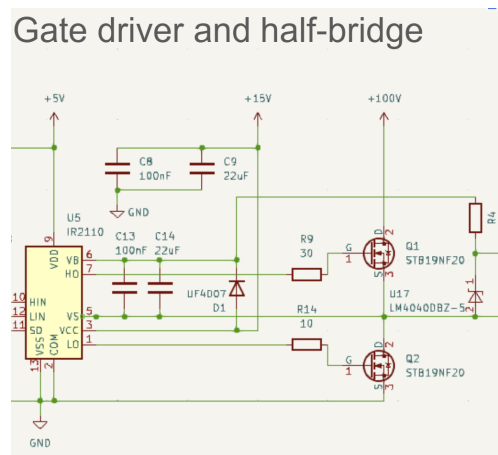


Figure 13: PCB of Gate Driver + Half-Bridge

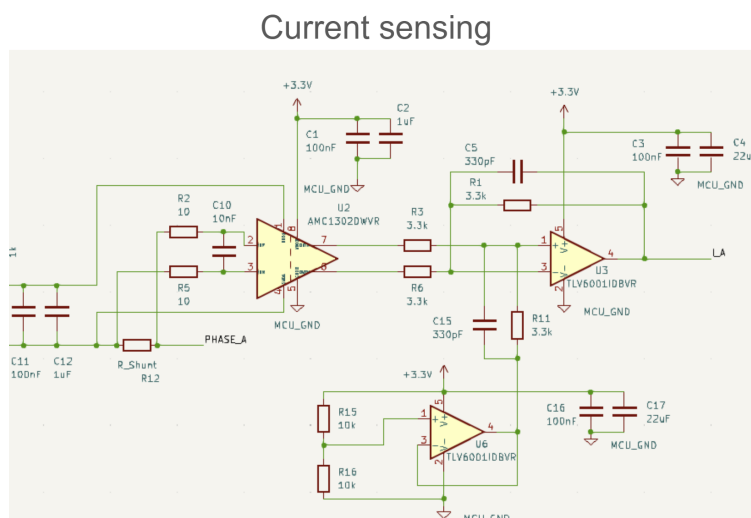


Figure 14: PCB of Current sensor

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